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Citrus Greening (Huanglongbing) Disease and their Effective Management Practices

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Common Name

Citrus greening, citrus huanglongbing (greening) disease.

Disease Status

Greening disease is known to happen in Andhra Pradesh, Karnataka, Punjab, Uttar Pradesh, Himachal Pradesh, Rajasthan, Maharashtra, Jammu, Bihar, Bengal and Sikkim. The disease is more extreme on sweet oranges than on corrosive lime, mandarin and grapefruits. This infection influences practically all citrus assortments independent of root stock. In India Mosambi sweet orange and Darjeeling orange *(Citrus reticulata)* are acceptable pointer plants for greening.

Causal Organism

Citrus greening disease (Yellow dragon disease or HLB) is a sickness of citrus brought about by a vector-transmitted microbe (1). The causative specialists are motile microbes, *Liberibacter* spp. The disease is vectored and transmitted by the Asian citrus psyllid, *Diaphorina citri,* and the African citrus psyllid, *Trioza erytreae,* otherwise called the two-spotted citrus psyllid (2). It has additionally been demonstrated to be joining transmissible. *Candidatus Liberibacter asiaticus* (Fastidious Phloem restricted Bacterium), commit gram negative bacterium.

Hosts

All species and varieties of citrus are affected, such as orange, grapefruit, mandarin, tangelo, kumquat, lemon, lime, pomelo, trifoliate orange and tangelo. Mock orange or orange jasmine *(Murraya paniculata)*, and the curry tree *(Bergera koenigii)* are also hosting.

Symptoms



(Source: researchgate.net)

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Affected trees are stunted with pronounced leaf and fruit drop. Varied chlorotic patterns on leaves are noticed which are persistent and cannot be corrected by zinc sprays. Reduction in leaf size is common. Many affected leaves show small circular green islands within the chlorotic areas. Heavy leaf fall occurs with the onset of summer.

Often new flush may come out and leaves formed are short, upright and chlorotic with green veins or with green blotches on the leaves. Twig die-back occurs. Some branches in a tree exhibit severe symptoms whereas others in the same tree are apparently normal. Fruits show reduction in size, lopsided growth and oblique columella. The rind surface exposed to sun appears yellow whereas the remaining portion remains dull green. Diseased fruits are valueless owing to small size, distortion, low in juice and soluble solids, high in acid and insipid taste. Seeds are poorly developed, dark coloured and aborted.

Predisposing Factors

1. Mode of spread: The disease is transmitted through tainted bud wood and citrus psylla, *Diaphorina citri*. The infection likewise transmits from citrus to Periwinkle *(Catharanthus roseus)* through dodder.

2. Management:

a. Certain antibiotics, specifically streptomycin and oxytetracycline, may be effective in the fight against citrus greening disease and have been used in the United States but have been banned in Brazil and the European Union (3).

b. Natural Enemies: The nymphs are killed by parasitic wasps and predators including ladybeetles - larvae and adults - syrphid fly larvae, and pirate bugs. In California, the eulophid wasp, Tamarixia radiata, has been imported from the Punjab of Pakistan and released. Introduction of Tamarixia radiata has to be done with care, as there are hyperparasites that attack it. Research on Tamarixia radiata has also been done in Reunion and Mauritius.

c. Cultural Control: Cultural controls are important. Citrus should be monitored for huanglongbing and psyllids. It should be done to confirm that they are not present, or if they are present whether they are spreading. It is usual for the psyllid to be found first, and then 1-2 years later the disease appears. An IPM program against this disease requires (i) disease-free nursery plants, (ii) monitoring, and (iii) removal of infected trees (i.e., those with symptoms).

d. Check that plants supplied by commercial or government nurseries have been tested for huanglongbing bacteria and important viruses. The use of certified healthy trees is important in the management of the disease. It is also important that as part of the certified healthy tree program the nursery stocks are grown under psyllid-proof screenhouses.

e. Do not allow ornamental plants that are hosts of huanglongbing and psyllid, for instance, orange jasmine *(Murraya paniculata)* and curry leaf *(Bergera koenigii)*, to be moved from areas where the disease and/or insects occur.

f. It is important to monitor citrus for disease symptoms and for the presence of the psyllid. Monitoring should be done by farmers (preferably monthly if in a disease area, otherwise every 3 months), and also by quarantine authorities using yellow sticky traps placed in the tree canopy. Citrus growers should be trained how to do this. Early detection of the disease and the psyllid is very important. If diseased trees are found they should be removed, and burnt, but before removal they should be sprayed with the chemicals listed below to prevent spread of the psyllids.

3. Chemical Control: Mostly two insecticides are applied in areas where the psyllid is found:

a. Spray trees with a pyrethroid (e.g., lambda-cyhalothrin) insecticide to kill the adults and nymphs. Malathion can also be used.

b. Spray the ground around trees with a systemic product (e.g., imidacloprid) so that it is taken up by the roots to kill nymphs in the folds of leaves in the canopy. Note, do not use imidacloprid when trees are flowering as it is toxic to bees.

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c. As an alternative to synthetic products, use "soft" insecticides, e.g., horticultural spray oil, neem or insecticidal soap

d. These treatments give protection for 9-12 months.

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